TECHNICAL PROJECT REPORT

Coffee Shop Network Infrastructure Project

Cisco Packet Tracer | VLAN Segmentation, Inter-VLAN Routing, DHCP, ACL, Static Routing and NAT

Simulation platform: Cisco Packet Tracer

Devices: Cisco 2911 router, Cisco 3560 multilayer switch, simulated ISP router and server

Scope: Lab simulation. No real public Internet connectivity was established.

1. Project Overview and Objectives

This project is a segmented coffee shop network designed and implemented in Cisco Packet Tracer. The network separates Office, Point of Sale (POS), Guest Wi-Fi, and Network Management traffic while providing internal connectivity and access to a simulated external server.

1.1 Objectives

- Configure VLANs and access ports on a Cisco 3560 multilayer switch.
- Implement inter-VLAN routing using Switch Virtual Interfaces (SVIs).
- Provide DHCP services from the multilayer switch.
- Configure static routing between the multilayer switch, the CoffeeShop Router, and the ISP Router.
- Configure Guest Wi-Fi and restrict Guest access to internal networks with an extended ACL.
- Implement NAT overload (PAT) so internal networks share the router's WAN address.
- Verify connectivity, DHCP allocation, VLAN isolation, routing, and NAT operation.
- Secure unused switch ports and save device configurations.

1.2 Scope of the Simulation

> **Simulation notice**
> The external server at **198.51.100.10** is a Server-PT device simulated in Packet Tracer. The WAN (203.0.113.0/30) and server (198.51.100.0/24) subnets are documentation-only address ranges. **Real public Internet connectivity was not established.**

2. Network Devices and Technologies

2.1 Devices

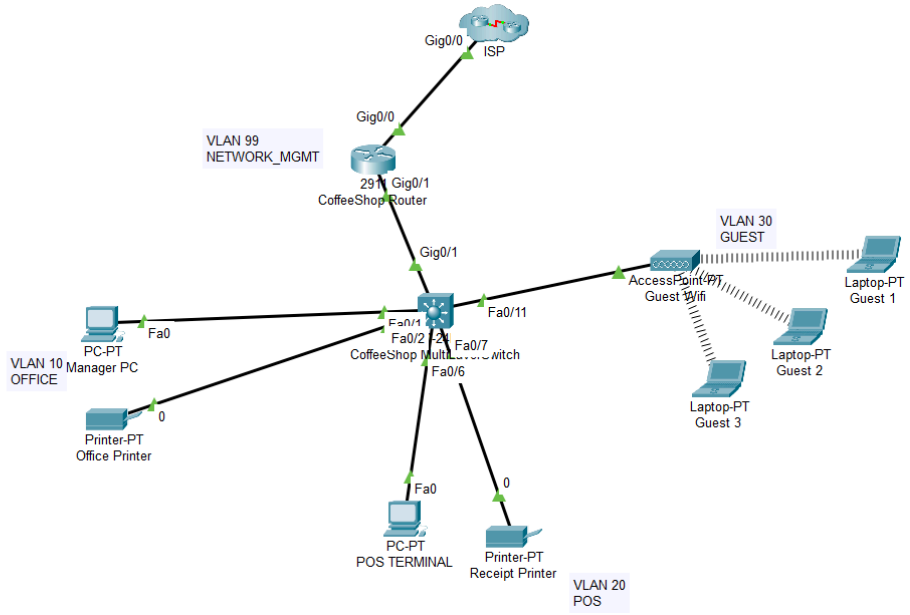
Device	Role
Cisco 2911 CoffeeShop Router	WAN connectivity, static routing, NAT overload
Cisco 3560 CoffeeShop_SW	VLANs, inter-VLAN routing, DHCP, Guest ACL
ISP Router	Simulated ISP and external server connectivity
Server-PT	Simulated Internet destination
Manager PC	Office network client
Office Printer	Office printing
POS Terminal	POS network client
Receipt Printer	POS printing
Wireless Access Point	Guest Wi-Fi connectivity
Guest Laptop	Guest network client

2.2 Technologies

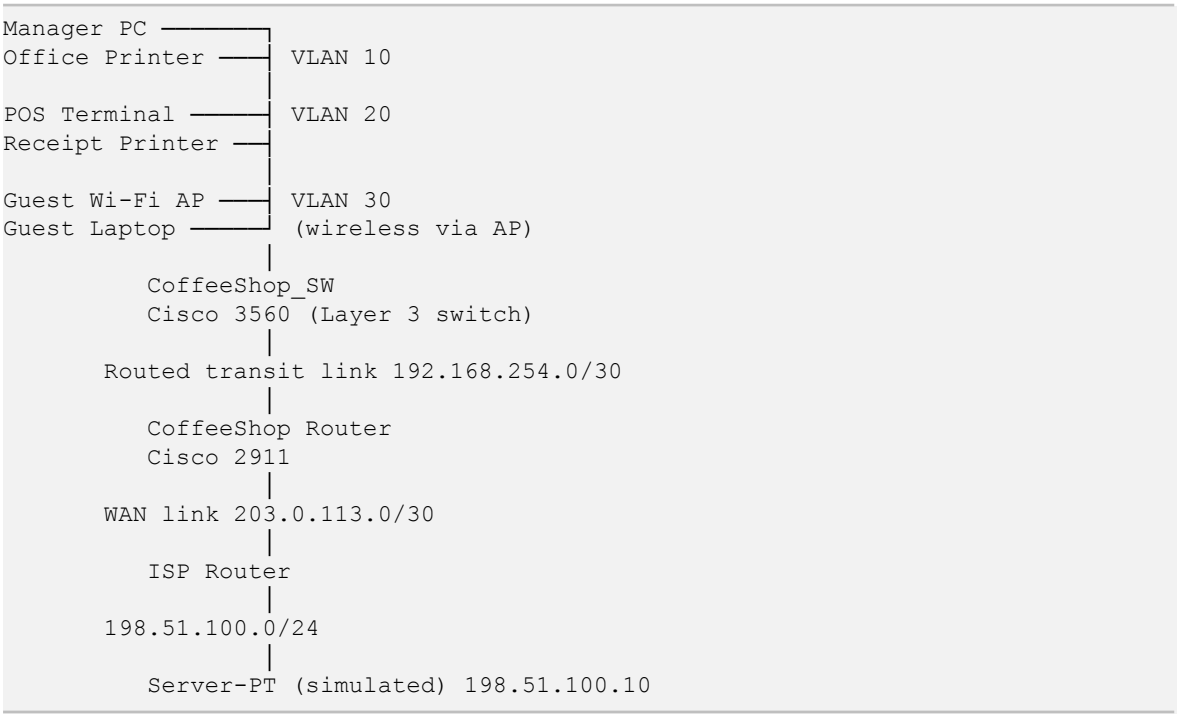
- VLAN segmentation and access port configuration
- Layer 3 switching with SVIs (inter-VLAN routing)
- DHCP server pools with excluded address ranges
- Static and default routing
- Extended named ACL for Guest restrictions
- NAT overload (PAT) with a standard ACL
- PortFast, BPDU Guard, and unused-port hardening

3. Network Topology

The multilayer switch handles internal VLAN routing and DHCP. The CoffeeShop Router connects the internal networks to the simulated ISP and performs NAT overload. The Guest ACL restricts Guest access to the Office, POS, and Management subnets.



3.1 Logical Traffic Flow



3.2 Traffic Flow Summary

- **Inter-VLAN traffic:** clients send traffic to their VLAN SVI on CoffeeShop_SW, which routes it to the destination VLAN.
- **Outbound traffic to the simulated server:** client, VLAN SVI, default route via 192.168.254.1, CoffeeShop Router (NAT overload to 203.0.113.2), ISP Router, Server-PT.
- **Return traffic:** the server replies to 203.0.113.2; the router reverses the translation and routes the packet to the switch using its static routes.
- **Guest traffic:** Guest clients obtain DHCP addresses and reach their own gateway; traffic to Office, POS, and Management subnets is denied by the ACL on the VLAN 30 SVI.

4. IP Addressing and VLAN Design

4.1 VLAN Summary

VLAN ID	Name	Subnet	Gateway	DHCP
10	OFFICE	192.168.10.0/24	192.168.10.1	Yes
20	POS	192.168.20.0/24	192.168.20.1	Yes
30	GUEST	192.168.30.0/24	192.168.30.1	Yes
99	NET_MGMT	192.168.99.0/24	192.168.99.1	No
999	PARKING	No IP	N/A	No

VLAN segmentation creates separate Layer 2 broadcast domains. Inter-VLAN communication is performed by the multilayer switch through its Layer 3 interfaces.

4.2 Internal Addressing

Device / Network	Interface	VLAN	IP address	Assignment
CoffeeShop_SW SVI	VLAN 10	10	192.168.10.1	Static (gateway)
Manager PC	Fa0/1	10	192.168.10.50*	DHCP
Office Printer	Fa0/2	10	192.168.10.3	Static
Office reserved range	N/A	10	192.168.10.4–49	Reserved for future devices
CoffeeShop_SW SVI	VLAN 20	20	192.168.20.1	Static (gateway)
POS Terminal	Fa0/6	20	192.168.20.50*	DHCP
Receipt Printer	Fa0/7	20	192.168.20.3	Static
POS reserved range	N/A	20	192.168.20.4–49	Reserved for future devices
CoffeeShop_SW SVI	VLAN 30	30	192.168.30.1	Static (gateway)
Guest Wi-Fi clients	Wireless	30	192.168.30.50–254	DHCP
CoffeeShop_SW SVI	VLAN 99	99	192.168.99.1	Static (gateway)
Future management devices	To be assigned	99	192.168.99.10–12	Reserved

* The DHCP pools begin at .50 because .1–.49 are excluded. Actual leases may change if leases are renewed or the configuration changes. The wireless AP's own management address, if configured, is separate from the Guest clients' DHCP addresses.

4.3 Transit and Simulated ISP Networks

Network	Device / Interface	IP address	Purpose
192.168.254.0/30 (255.255.255.252)	CoffeeShop Router G0/1	192.168.254.1	Transit gateway, NAT inside
	CoffeeShop_SW Gi0/1 (routed)	192.168.254.2	Routed switch uplink
203.0.113.0/30 (255.255.255.252)	ISP Router G0/0	203.0.113.1	ISP next hop
	CoffeeShop Router G0/0	203.0.113.2	WAN interface, NAT outside
198.51.100.0/24 (255.255.255.0)	ISP Router G0/1	198.51.100.1	Server gateway
	Server-PT NIC	198.51.100.10	Simulated external server

5. Switch Port Assignments

Switch port	VLAN	Connected device / use
Fa0/1	10	Manager PC
Fa0/2	10	Office Printer
Fa0/3–Fa0/5	10	Office expansion ports

Switch port	VLAN	Connected device / use
Fa0/6	20	POS Terminal
Fa0/7	20	Receipt Printer
Fa0/8–Fa0/10	20	POS expansion ports
Fa0/11	30	Guest Wi-Fi AP
Fa0/12–Fa0/24	999	Unused, shut down
Gi0/1	Routed	Link to CoffeeShop Router G0/1
Gi0/2	999	Unused, shut down

No access ports are assigned to VLAN 99 in this design. VLAN 99 is reserved for future management devices (see Section 14 and Section 15).

6. Multilayer Switch Configuration

All commands in this section are entered on CoffeeShop_SW (Cisco 3560).

6.1 VLANs and Layer 3 Routing

```
CoffeeShop_SW(config)#do show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   OFFICE                           active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5
20   POS                              active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
30   GUEST                            active    Fa0/11
99   NET_MGMT                         active
999  PARKING                          active    Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
```

```
vlan 10
 name OFFICE
vlan 20
 name POS
vlan 30
 name GUEST
vlan 99
 name NET_MGMT
vlan 999
 name PARKING

ip routing
```

```
!
ip routing
!
```

The ip routing command enables the switch to route traffic between VLANs.

6.2 VLAN Gateways (SVIs)

```
interface Vlan10
 description OFFICE_GATEAWAY
 mac-address 000c.cf10.0901
 ip address 192.168.10.1 255.255.255.0
!
interface Vlan20
 description POS_GATEWAY
 mac-address 000c.cf10.0902
 ip address 192.168.20.1 255.255.255.0
!
interface Vlan30
 description GUEST_GATEWAY
 mac-address 000c.cf10.0903
 ip address 192.168.30.1 255.255.255.0
 ip access-group GUEST_RESTRICTIONS in
!
interface Vlan99
 description NETWORK_MANAGEMENT
 mac-address 000c.cf10.0904
 ip address 192.168.99.1 255.255.255.0
```

```
interface vlan 10
 description OFFICE_GATEWAY
 ip address 192.168.10.1 255.255.255.0
 no shutdown
interface vlan 20
 description POS_GATEWAY
 ip address 192.168.20.1 255.255.255.0
 no shutdown
interface vlan 30
 description GUEST_GATEWAY
 ip address 192.168.30.1 255.255.255.0
 no shutdown
interface vlan 99
 description MANAGEMENT_GATEWAY
 ip address 192.168.99.1 255.255.255.0
 no shutdown
```

Each SVI is the default gateway for its VLAN. An SVI normally becomes operational only when its VLAN exists and has an active Layer 2 port, subject to SVI autostate behavior.

6.3 Access Ports

```
interface FastEthernet0/1
 description OFFICE_DEVICES
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 description OFFICE_DEVICES
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 description OFFICE_DEVICES
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 description OFFICE_DEVICES
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/5
 description OFFICE_DEVICES
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
```

```
interface FastEthernet0/6
 description POS_DEVICES
 switchport access vlan 20
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/7
 description POS_DEVICES
 switchport access vlan 20
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/8
 description POS_DEVICES
 switchport access vlan 20
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/9
 description POS_DEVICES
 switchport access vlan 20
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/10
 description POS_DEVICES
 switchport access vlan 20
 switchport mode access
 spanning-tree portfast
```

```
interface FastEthernet0/11
 description GUEST_WIFI
 switchport access vlan 30
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
```

```
interface range fastEthernet 0/1 - 5
 description OFFICE_DEVICES
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable

interface range fastEthernet 0/6 - 10
 description POS_DEVICES
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 spanning-tree bpduguard enable

interface fastEthernet 0/11
 description GUEST_WIFI
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 spanning-tree bpduguard enable
```

Fa0/11 places the AP's wired connection in VLAN 30; Guest wireless clients are bridged into that same network.

6.4 Unused Ports

```
interface FastEthernet0/12
 description UNUSED_PORTS
 switchport access vlan 999
 switchport mode access
 shutdown
!
interface FastEthernet0/13
 description UNUSED_PORTS
 switchport access vlan 999
 switchport mode access
 shutdown
!
interface FastEthernet0/14
 description UNUSED_PORTS
 switchport access vlan 999
 switchport mode access
 shutdown
!
interface FastEthernet0/15
 description UNUSED_PORTS
 switchport access vlan 999
 switchport mode access
 shutdown
!
interface FastEthernet0/16
 description UNUSED_PORTS
 switchport access vlan 999
 switchport mode access
 shutdown
```

```
interface range fastEthernet 0/12 - 24
 description UNUSED_PORTS
 switchport mode access
 switchport access vlan 999
 shutdown

interface gigabitEthernet 0/2
 description UNUSED_UPLINK
 switchport mode access
 switchport access vlan 999
 shutdown
```

Unused ports are placed in the parking VLAN and administratively shut down to reduce the risk of unauthorized devices connecting. VLAN 999 has no IP address or gateway.

6.5 Routed Uplink and Default Route

```
interface GigabitEthernet0/1
 description LINK_TO_COFFEESHOP_ROUTER
 no switchport
 ip address 192.168.254.2 255.255.255.252
 duplex auto
 speed auto
```

```
ip route 0.0.0.0 0.0.0.0 192.168.254.1
```

```
interface gigabitEthernet 0/1
 description LINK_TO_COFFEESHOP_ROUTER
 no switchport
 ip address 192.168.254.2 255.255.255.252
 no shutdown

ip route 0.0.0.0 0.0.0.0 192.168.254.1
```

The no switchport command converts Gi0/1 into a routed Layer 3 interface. The default route sends traffic for destinations not in the switch's routing table to the CoffeeShop Router.

7. DHCP Configuration

CoffeeShop_SW provides DHCP for the Office, POS, and Guest networks. Addresses .1–.49 are excluded in each subnet to protect the gateway and the planned static address range.

```
hostname CoffeeShop_SW
!
!
!
ip dhcp excluded-address 192.168.10.1 192.168.10.49
ip dhcp excluded-address 192.168.20.1 192.168.20.49
ip dhcp excluded-address 192.168.30.1 192.168.30.49
!
ip dhcp pool OFFICE
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool POS
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
ip dhcp pool GUEST
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
```

```
ip dhcp excluded-address 192.168.10.1 192.168.10.49
ip dhcp excluded-address 192.168.20.1 192.168.20.49
ip dhcp excluded-address 192.168.30.1 192.168.30.49

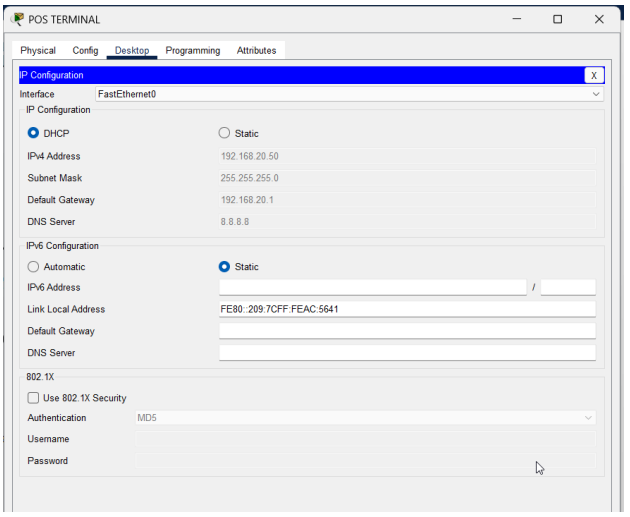
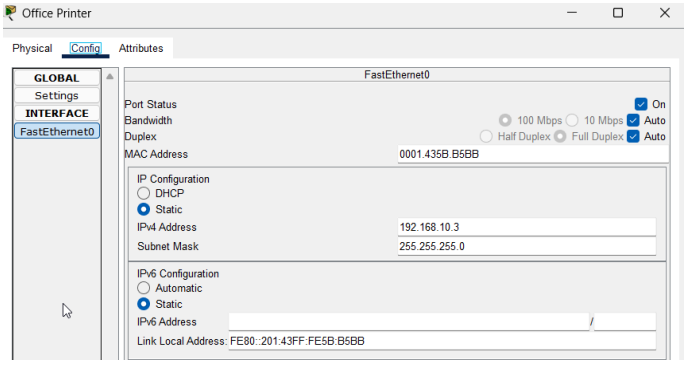
ip dhcp pool OFFICE
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool POS
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
ip dhcp pool GUEST
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
```

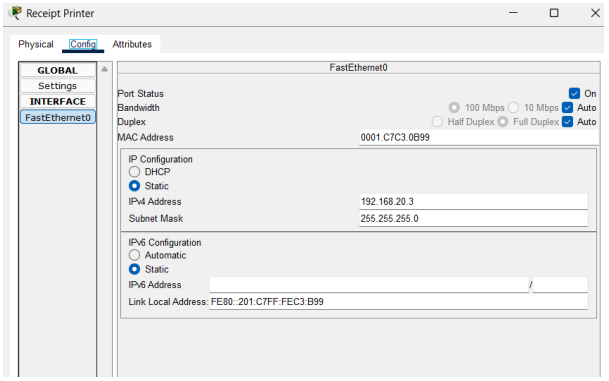
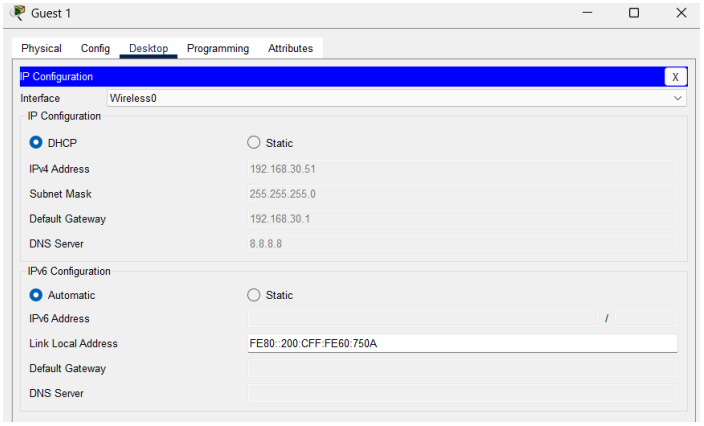
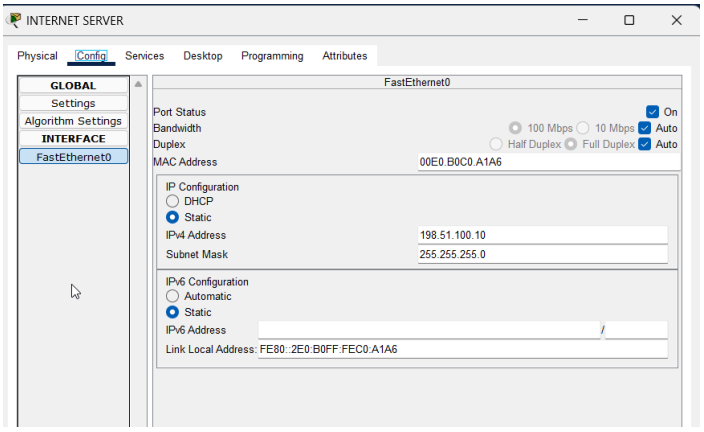
The DNS server 8.8.8.8 is only a DHCP-delivered setting. It is not proof that the simulated network can reach that address, and DNS resolution was not part of this project.

8. End Device Configuration

Device	Addressing	Expected configuration

Manager PC	DHCP	192.168.10.50 (or another lease), mask 255.255.255.0, gateway 192.168.10.1
		`hostname CoffeeShop_SW`
		`!`
		`!`
		`!`
		`ip dhcp excluded-address 192.168.10.1 192.168.10.49`
		`ip dhcp excluded-address 192.168.20.1 192.168.20.49`
		`ip dhcp excluded-address 192.168.30.1 192.168.30.49`
		`!`
		`ip dhcp pool OFFICE`
		` network 192.168.10.0 255.255.255.0`
		` default-router 192.168.10.1`
		` dns-server 8.8.8.8`
		`ip dhcp pool POS`
		` network 192.168.20.0 255.255.255.0`
		` default-router 192.168.20.1`
		` dns-server 8.8.8.8`
		`ip dhcp pool GUEST`
		` network 192.168.30.0 255.255.255.0`
		` default-router 192.168.30.1`
		` dns-server 8.8.8.8`

POS Terminal	DHCP	192.168.20.50 (or another lease), mask 255.255.255.0, gateway 192.168.20.1 
Office Printer	Static	192.168.10.3, mask 255.255.255.0, gateway 192.168.10.1 

Device	Addressing	Expected configuration
Receipt Printer	Static	192.168.20.3, mask 255.255.255.0, gateway 192.168.20.1 
Guest Laptop (wireless)	DHCP	192.168.30.50 (or another lease), mask 255.255.255.0, gateway 192.168.30.1 
Server-PT	Static	198.51.100.10, mask 255.255.255.0, gateway 198.51.100.1 

The wireless AP was configured with an SSID and wireless security in Packet Tracer and connected to Fa0/11 (VLAN 30). The wireless password is a lab value and is intentionally not published in this report. The AP's own management address, if configured, should be documented separately from Guest client addresses.

9. Router and ISP Configuration

9.1 CoffeeShop Router (Cisco 2911)

```
interface GigabitEthernet0/1
 description ROUTER_LINK_TO_MULTILAYER_SWITCH
 ip address 192.168.254.1 255.255.255.252
 ip nat inside
 duplex auto
 speed auto

interface GigabitEthernet0/0
 description WAN_TO_ISP
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 duplex auto
 speed auto
ip classless
ip route 192.168.10.0 255.255.255.0 192.168.254.2
ip route 192.168.20.0 255.255.255.0 192.168.254.2
ip route 192.168.30.0 255.255.255.0 192.168.254.2
ip route 192.168.99.0 255.255.255.0 192.168.254.2
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

```
interface gigabitEthernet 0/1
 description ROUTER_LINK_TO_MULTILAYER_SWITCH
 ip address 192.168.254.1 255.255.255.252
 ip nat inside
 no shutdown

interface gigabitEthernet 0/0
 description WAN_TO_ISP
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 no shutdown

ip route 192.168.10.0 255.255.255.0 192.168.254.2
ip route 192.168.20.0 255.255.255.0 192.168.254.2
ip route 192.168.30.0 255.255.255.0 192.168.254.2
ip route 192.168.99.0 255.255.255.0 192.168.254.2
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

G0/1 faces the multilayer switch (NAT inside) and G0/0 faces the ISP (NAT outside). The four static routes direct traffic for each internal VLAN to the switch; the default route sends everything else to the ISP Router.

9.2 ISP Router (Simulated)

```
interface GigabitEthernet0/0
 description LINK_TO_COFFEESHOP
 ip address 203.0.113.1 255.255.255.252
 duplex auto
 speed auto
```

```
interface GigabitEthernet0/1
 description LINK_TO_INTERNET_SERVER
 ip address 198.51.100.1 255.255.255.0
 duplex auto
 speed auto
```

```
interface gigabitEthernet 0/0
 description LINK_TO_COFFEESHOP
 ip address 203.0.113.1 255.255.255.252
 no shutdown

interface gigabitEthernet 0/1
 description LINK_TO_INTERNET_SERVER
 ip address 198.51.100.1 255.255.255.0
 no shutdown
```

The ISP Router has directly connected routes for 203.0.113.0/30 and 198.51.100.0/24. A static route back to the internal networks is not required for the NAT test, because the server replies to the translated address 203.0.113.2, which is on the directly connected WAN subnet.

10. NAT Overload Configuration

NAT overload (PAT) allows multiple internal addresses to share the CoffeeShop Router's WAN address, 203.0.113.2, when communicating with external destinations.

```
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.99.0 0.0.0.255
!
!
ip nat inside source list 1 interface GigabitEthernet0/0 overload
ip classless
ip route 192.168.10.0 255.255.255.0 192.168.254.2
ip route 192.168.20.0 255.255.255.0 192.168.254.2
ip route 192.168.30.0 255.255.255.0 192.168.254.2
ip route 192.168.99.0 255.255.255.0 192.168.254.2
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
ip flow-export version 9
!
!
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.99.0 0.0.0.255
!
```

```
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.99.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet 0/0 overload
```

Standard ACL 1 only identifies which source subnets are eligible for translation. It does not provide Guest isolation; that is handled by the separate extended ACL on VLAN 30.

Result: NAT translations were observed for traffic from the Office PC to 198.51.100.10, using the global address 203.0.113.2 (verified with show ip nat translations and show ip nat statistics).

11. Guest Network Security and ACL

11.1 Intended Guest Policy

- Obtain IP configuration through DHCP.
- Reach the Guest default gateway.
- Reach permitted external destinations.
- Be blocked from the Office, POS, and Management subnets.

11.2 Extended ACL on CoffeeShop_SW

```
!
ip access-list extended GUEST_RESTRICTIONS
 permit udp any eq bootpc any eq bootps
 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.99.0 0.0.0.255
 permit ip 192.168.30.0 0.0.0.255 any
!
!
interface Vlan30
 description GUEST_GATEWAY
 mac-address 000c.cf10.0903
 ip address 192.168.30.1 255.255.255.0
 ip access-group GUEST_RESTRICTIONS in
!
```

```
ip access-list extended GUEST_RESTRICTIONS
 permit udp any eq bootpc any eq bootps
 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.99.0 0.0.0.255
 permit ip 192.168.30.0 0.0.0.255 any

interface vlan 30
 ip access-group GUEST_RESTRICTIONS in
```

11.3 Rule Order

The first rule permits DHCP client requests (UDP source port 68, bootpc, to UDP destination port 67, bootps). It must appear before the deny rules because ACLs are evaluated top to bottom and processing stops at the first match. The rule is deliberately narrow so it does not open access to internal subnets. A broad permit placed before the deny rules would allow access to internal networks.

11.4 Expected Guest Access Policy

Destination	Expected result
Guest gateway 192.168.30.1	Allowed
Office subnet 192.168.10.0/24	Blocked
POS subnet 192.168.20.0/24	Blocked

Destination	Expected result
Management subnet 192.168.99.0/24	Blocked
Simulated server 198.51.100.10	Allowed, provided routing and NAT work

The final permit allows other IPv4 destinations not matched by the earlier deny rules. It does not guarantee that those destinations are reachable or that the required services exist.

12. Network Verification and Test Results

Results are separated into confirmed tests (completed in the lab) and expected results (design intent that has not been recorded as a completed test).

12.1 Confirmed Results

Test	Result
Inter-VLAN routing	Confirmed
Office DHCP	Confirmed
POS DHCP	Confirmed
Guest DHCP after ACL fix	Confirmed working
Guest isolation from Office subnet	Confirmed (tested with a static Guest address, before the DHCP exception was added)
Guest isolation from POS subnet	Confirmed (same conditions)
Guest isolation from Management subnet	Confirmed (same conditions)
Office PC to simulated server 198.51.100.10	Confirmed working
NAT translations (global address 203.0.113.2)	Observed for Office PC traffic to the server

Guest isolation (static address) and Guest DHCP (after the ACL fix) were confirmed as separate checks.

12.2 Expected or Not Yet Recorded

Item	Status
VLAN creation and port assignments	Configured; final state to be verified with show vlan brief
Guest laptop reaching the simulated server	Expected to succeed if routing and NAT work; not recorded as a completed test
Printer reachability (Office and Receipt printers)	Individual results not recorded. A ping verifies IP connectivity only, not printing services
Vlan99 interface state	No access port is assigned to VLAN 99, so the SVI state depends on autostate; up/up is not confirmed
Configuration saved on all devices	Complete only after copy running-config startup-config is run on each device

12.3 Verification Commands

! **CoffeeShop_SW**

show vlan brief

```
CoffeeShop_SW#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   OFFICE                           active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5
20   POS                              active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
30   GUEST                            active    Fa0/11
99   NET_MGMT                         active
999  PARKING                          active    Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

show ip interface brief

```
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
FastEthernet0/3        unassigned      YES unset  down                  down
FastEthernet0/4        unassigned      YES unset  down                  down
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        unassigned      YES unset  up                    up
FastEthernet0/7        unassigned      YES unset  up                    up
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  up                    up
FastEthernet0/12       unassigned      YES unset  administratively down down
FastEthernet0/13       unassigned      YES unset  administratively down down
FastEthernet0/14       unassigned      YES unset  administratively down down
FastEthernet0/15       unassigned      YES unset  administratively down down
FastEthernet0/16       unassigned      YES unset  administratively down down
FastEthernet0/17       unassigned      YES unset  administratively down down
FastEthernet0/18       unassigned      YES unset  administratively down down
FastEthernet0/19       unassigned      YES unset  administratively down down
FastEthernet0/20       unassigned      YES unset  administratively down down
FastEthernet0/21       unassigned      YES unset  administratively down down
FastEthernet0/22       unassigned      YES unset  administratively down down
FastEthernet0/23       unassigned      YES unset  administratively down down
FastEthernet0/24       unassigned      YES unset  administratively down down
GigabitEthernet0/1     192.168.254.2   YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.20.1    YES manual up                    up
Vlan30                 192.168.30.1    YES manual up                    up
Vlan99                 192.168.99.1    YES manual up                    down
CoffeeShop_SW#
```

show ip route

```
CoffeeShop_SW#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.254.1 to network 0.0.0.0

C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
     192.168.254.0/30 is subnetted, 1 subnets
C       192.168.254.0 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 192.168.254.1
```

show ip dhcp pool

```
Pool OFFICE :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.1         192.168.10.1     - 192.168.10.254    1    / 3     / 254

Pool POS :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.20.1         192.168.20.1     - 192.168.20.254    1    / 3     / 254

Pool GUEST :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 3
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
```

```
1 subnet is currently in the pool
Current index        IP address range                    Leased/Excluded/Total
192.168.30.1         192.168.30.1     - 192.168.30.254    3    / 3     / 254
```

show ip dhcp binding

```
CoffeeShop_SW# show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.50    0007.ECC7.11DC           --                     Automatic
192.168.20.50    0009.7CAC.5641           --                     Automatic
192.168.30.51    0000.0C60.750A           --                     Automatic
192.168.30.53    0050.0F22.11A8           --                     Automatic
192.168.30.55    0004.9A10.78CA           --                     Automatic
```

! **CoffeeShop Router**

show ip interface brief

```
COFFEESHOP_ROUTER#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     203.0.113.2     YES manual up                    up
GigabitEthernet0/1     192.168.254.1   YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
COFFEESHOP ROUTER#
```

show ip route

```
COFFEESHOP_ROUTER#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

S    192.168.10.0/24 [1/0] via 192.168.254.2
S    192.168.20.0/24 [1/0] via 192.168.254.2
S    192.168.30.0/24 [1/0] via 192.168.254.2
S    192.168.99.0/24 [1/0] via 192.168.254.2
     192.168.254.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.254.0/30 is directly connected, GigabitEthernet0/1
L       192.168.254.1/32 is directly connected, GigabitEthernet0/1
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0
L       203.0.113.2/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 203.0.113.1

COFFEESHOP ROUTER#
```

```
show ip nat translations
show ip nat statistics
```

```
COFFEESHOP_ROUTER#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.2:29    192.168.10.50:29   203.0.113.1:29     203.0.113.1:29
icmp 203.0.113.2:30    192.168.10.50:30   198.51.100.10:30   198.51.100.10:30
icmp 203.0.113.2:38    192.168.30.51:38   203.0.113.1:38     203.0.113.1:38
icmp 203.0.113.2:39    192.168.30.51:39   198.51.100.10:39   198.51.100.10:39

COFFEESHOP_ROUTER#show ip nat statistic
Total translations: 1 (0 static, 1 dynamic, 1 extended)
Outside Interfaces: GigabitEthernet0/0
Inside Interfaces: GigabitEthernet0/1
Hits: 21  Misses: 33
Expired translations: 22
Dynamic mappings:
COFFEESHOP_ROUTER#
```

```
! ISP Router
show ip interface brief
```

```
ISP_ROUTER#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     203.0.113.1     YES manual up                    up
GigabitEthernet0/1     198.51.100.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
show ip route
```

```
ISP_ROUTER#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/24 is directly connected, GigabitEthernet0/1
L       198.51.100.1/32 is directly connected, GigabitEthernet0/1
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0
L       203.0.113.1/32 is directly connected, GigabitEthernet0/0
```

Client-side checks used ipconfig and ping from the Packet Tracer command prompt (for example ping 192.168.20.1 from the Manager PC to test inter-VLAN routing, and ping 198.51.100.10 to test the path to the simulated server).

Expected routing output: the switch shows connected VLAN and transit routes plus a default route via 192.168.254.1; the CoffeeShop Router shows the internal subnets via 192.168.254.2, connected transit and WAN subnets, and a default route via 203.0.113.1; the ISP Router shows connected routes for 203.0.113.0/30 and 198.51.100.0/24.

13. Troubleshooting Case Study: Guest DHCP Failure

Stage	Details
Symptom	The Guest laptop could not obtain an address from 192.168.30.0/24 and received an APIPA address.
Investigation	The Guest ACL was applied inbound on the VLAN 30 SVI. DHCP requests were not being permitted before the network restrictions were evaluated, so they were blocked.
Resolution	Added a DHCP permit rule at the top of GUEST_RESTRICTIONS, ahead of the deny statements.
Outcome	Guest DHCP began working after the rule was added.

```
permit udp any eq bootpc any eq bootps
!
ip access-list extended GUEST_RESTRICTIONS
 permit udp any eq bootpc any eq bootps
 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.99.0 0.0.0.255
 permit ip 192.168.30.0 0.0.0.255 any
!
```

Lessons learned: account for essential network services such as DHCP when applying ACLs; rule order matters because traffic is evaluated top to bottom and processing stops at the first match; keep exceptions narrow.

14. Security Measures

14.1 Implemented Measures

Measure	Description
VLAN segmentation	Office, POS, Guest, and Management networks are separate subnets and broadcast domains
Guest network isolation	Extended ACL blocks Guest traffic to the Office, POS, and Management subnets
DHCP exception	A specific permit allows Guest DHCP requests before the deny rules
Unused port protection	Unused switch ports are placed in VLAN 999 and shut down
Access port protection	PortFast and BPDU Guard on end-device access ports
Network address translation	NAT overload translates eligible internal traffic to the WAN interface address

14.2 Security Limitations

- The Guest ACL isolates Guest clients from the listed internal subnets only. It is not a complete firewall policy and does not prevent Office or POS devices from initiating traffic toward other networks.
- The ACL denies the Office, POS, and Management subnets specifically; it does not explicitly deny other internal addresses such as the 192.168.254.0/30 transit network.
- VLAN 99 is designated for management, but no secure management access (for example SSH restrictions or management ACLs) is configured.
- NAT overload is address translation, not a security policy.

15. Project Limitations and Future Improvements

15.1 Limitations

- The project exists only in Packet Tracer; the ISP and server are simulated and do not provide real Internet connectivity.
- No DNS server is configured; 8.8.8.8 is only the value distributed by DHCP.

- Secure device management access is not configured.
- Printer service functionality (as opposed to IP reachability) is not demonstrated.
- No access port is assigned to VLAN 99.

15.2 Future Improvements

- Configure and document secure management access to network devices.
- Define an ACL policy that explicitly states which services each VLAN may access.
- Add a dedicated internal DNS server if name resolution is required in the simulation.
- Document AP management addressing and wireless security settings separately from Guest client addressing.
- Add a configuration backup and recovery procedure.
- Expand the project with monitoring, logging, and additional network services.

16. Skills Demonstrated

- Cisco IOS configuration and troubleshooting
- VLAN design and segmentation
- Layer 3 switching and inter-VLAN routing
- DHCP pool configuration and address management
- Static routing and default route configuration
- Extended ACL design and rule-order troubleshooting
- NAT overload (PAT)
- Network connectivity testing and route verification
- Basic switch port security practices
- Technical documentation and evidence-based verification

17. Conclusion

A segmented coffee shop network was implemented in Cisco Packet Tracer with separate Office, POS, Guest, and Management subnets. Inter-VLAN routing, DHCP, Guest isolation, and connectivity to a simulated external server were configured and tested, and NAT translations were observed for Office traffic to that server.

A Guest DHCP failure caused by the Guest ACL was diagnosed and resolved by adding a DHCP permit rule before the deny rules. The project demonstrates practical skills in VLAN design, Layer 3 switching, DHCP, static routing, ACLs, NAT, and troubleshooting, while clearly documenting its limitations as a lab simulation.